MOHAMED AZIZ HADJAYED

Embedded Systems Engineer

borj louzir ,Ariana ,Tunisie
+216 99664652
mohamedaziz.hadjayed@enicar.ucar.tn
www.linkedin.com/in/mohamed-aziz-hadj-ayed-0995a31b7
www.github.com/aziz-hadjayed

PROFILE

Final-year Embedded Systems Engineering student at the National Engineering School of Carthage. Passionate about Edge AI and IoT, with strong skills in C/C++, Python, TensorFlow/TFLite, and deploying lightweight AI models deploy on STM32 and Raspberry Pi platforms. Currently seeking a Final-Year Project (PFE) in Edge AI, IoT, and intelligent embedded systems.

EDUCATION

National Engineering School of Carthage (ENICarthage), Ariana, Tunisia
Engineering Degree in Embedded Systems **Sep 2023 – 2026**

Preparatory Institute for Engineering Studies of Monastir, Monastir, Tunisia
Pre-Engineering Degree in Physics and Technologies, Rank: 90 / 630 **2021 – 2023**

PROFESSIONAL EXPERIENCE

E-commerce with Content Recommendation Engine, Mobelite,
Monastir, Tunisia **15 July – 9 sep (2025)**

- Designed a web prototype using Streamlit, leveraging JSON files for user data storage and management.
- Implemented a content-based recommendation engine using scikit-learn, pandas, and NumPy.

Full-stack Developer Intern, DRÄXLMAIER(METS), Sousse, Tunisie **15 July - 30 August (2024)**

- Developed a full-stack web application using React, NestJS, and MySQL.
- Implemented a REST API for process management.
- Designed an intuitive user interface for workflow visualization.

PERSONAL PROJECTS

ESP32-Based Pass-Box Automation (Sterilization Cycle & Safety)

- ESP32 embedded system: 7-stage sterilization cycle, door interlocking, I2C LCD, MQTT/HiveMQ communication, FreeRTOS programming
- Complete IoT architecture: Real-time Node-RED dashboard, automatic CSV logging, dual email alert system (emergency + analytical reports), remote control

Embedded Autonomous Navigation System Based on TensorFlow

- Developed an autonomous robot on Raspberry Pi 4 using TensorFlow 2.x (CNN) for line detection.
- Implemented embedded vision, navigation AI, and motor control.

Security Evaluation Platform for Embedded Systems

- Developed a security evaluation solution on STM32F411RE.
- Built a monitoring interface using Flutter.
- Performed software and hardware attacks (UART, SWD/JTAG, Bleichenbacher).

IoT Solution for Automated Greenhouse Management

- Developed an ESP32-based system integrating multiple sensors for greenhouse monitoring.
- Designed an Android application for remote control and real-time data visualization.

Distributed IoT Network (Java)

- Sensor nodes + gateway + monitoring simulation using TCP sockets.
- Critical alerts via UDP multicast + distributed mutual exclusion (Lamport)

Sécurisation d'un système IoT (Raspberry Pi 4 – FastAPI + Flutter)

- 3 attaques sur Raspberry Pi 4 (serveur FastAPI) : DoS, reverse d'une app Flutter, extraction du firmware (carte SD).
- Mesure de l'impact (disponibilité/latence) et proposition de mitigations.
- Identification des faiblesses (configs/secrets/endpoints) et recommandations de contre-mesures.

TECHNICAL SKILLS

- Skills : **Python, C/C++ , UART,USART,I2C,SPI,CAN,Stm32(HAL/CMSIS), Esp32(ESP-IDF), Raspberry Pi, TensorFlow, PlatformIO , FastAPI, MQTT, Flutter, Embadded Linux , FreeRTOS ,Yocto Project (Kirkstone 4.0 LTS) / Poky**
- Software :**VScode ,STM32CubeIDE ,STM32Cube.AI ,MATLAB ,KALI Linux ,LABview ,ModelSIM ,GitHub ,KiCad9.0 , Firebase, Node-Red**

LANGUAGES

Arabe :Native language | French : B2 Level | English : B1 Level